

Foglio 6 - Soluzioni,

ESERCIZIO 1

$$A = \{(x, y) \in \mathbb{R}^2 \mid x^{2/3} + y^{2/3} \leq 1\}.$$

i) Calcolare

$$\iint_{A \cap \{x \geq 0, y \geq 0\}} xy \, dx \, dy$$

ii) Calcolare

$$\iint_A xy \, dx \, dy$$

Soluzione

Da **Gauss - Green**:
$$\iint_{A \cap \{x \geq 0, y \geq 0\}} xy \, dx \, dy = \int \frac{x^2 y}{2} \, dy \Big|_{\partial(A \cap \{x \geq 0, y \geq 0\})} = (*)$$

$$\gamma(t) = (\cos^3 t, \sin^3 t), \quad t \in [0, \pi/2].$$

$$\dot{\gamma}(t) = (-3\cos^2 t \sin t, 3\sin^2 t \cos t)$$

i)
$$(*) = \frac{3}{2} \int_0^{\pi/2} \cos^7 t \sin^5 t \, dt$$

$\cos t = u$
 $du = -\sin t \, dt$

$$= -\frac{3}{2} \int_1^0 u^7 (1-u^2)^2 \, du$$
$$= \frac{3}{2} \int_0^1 (u^7 + u^5 - 2u^3) \, du = \frac{1}{80}$$

ii)
$$\iint_A xy \, dx \, dy = 0 \quad \text{per simmetria.}$$

ESERCIZIO 2

$$w = \left(\frac{x-1-y}{(x-1)^2+y^2} + \frac{y}{(x+1)^2+y^2} \right) dx + \left(\frac{x-1+y}{(x-1)^2+y^2} - \frac{x+1}{(x+1)^2+y^2} \right) dy$$

Calcolare $\int_\gamma w$, con $\gamma(t) = (2\cos t, \sin 2t)$, $t \in [0, 2\pi]$

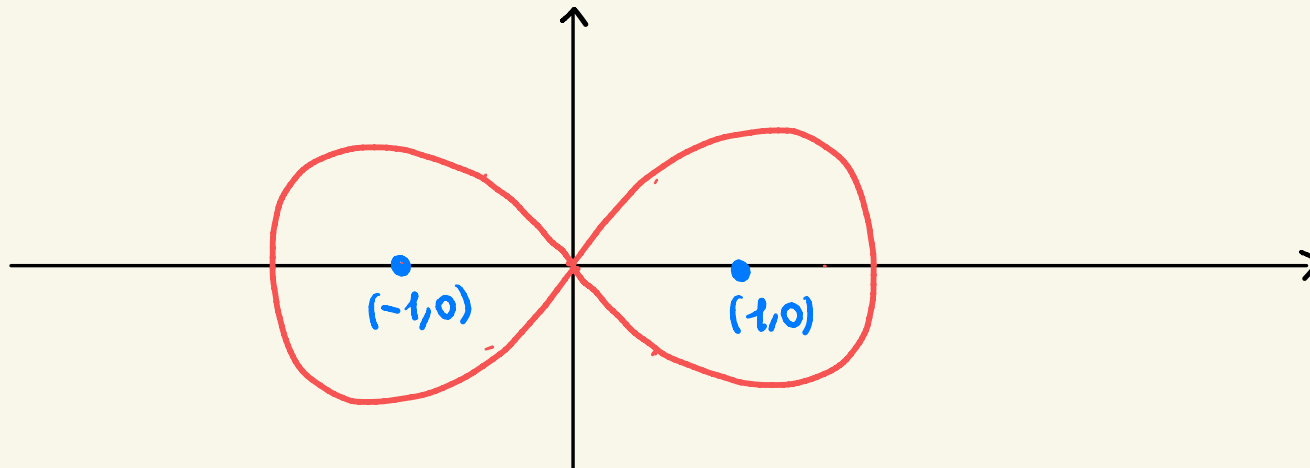
Soluzione

$$w = \hat{w}_{+1} + w_1 + w_2. \quad \text{dom}(w) = \mathbb{R}^2 \setminus \{(\pm 1, 0)\}$$

$$\text{con } \hat{w}_{+1} = \frac{(x-1)}{(x-1)^2+y^2} dx + \frac{y}{(x-1)^2+y^2} dy \quad \left[\text{chiusa su } \text{dom}(w_{+1}) > \text{dom}(w) - \text{visto a lezione} \right]$$

$$w_1 = \frac{(x-1)}{(x-1)^2+y^2} dx + \frac{y}{(x-1)^2+y^2} dy$$

$$w_2 = \frac{y}{(x+1)^2+y^2} dx - \frac{(x+1)}{(x+1)^2+y^2} dy$$



•) Studiamo w_1 .

$$\frac{\partial w_{1,x}}{\partial y} = \frac{\partial w_{1,y}}{\partial x} \longrightarrow w_1 \text{ chiusa in } \text{dom}(w_1) \supset \text{dom}(w)$$

È esatta? $\text{dom}(w_1)$ non è sempl. connesso. Proviamo a trovare un potenziale.

$$U_1(x,y) = \int w_{1,x}(x,y) dx = \frac{1}{2} \log((x-1)^2 + y^2) + C(y).$$

$$\frac{\partial U_1}{\partial y} = \frac{y}{(x-1)^2 + y^2} + C'(y) \stackrel{\text{INPONDO}}{=} w_{1,y}$$

Scegliendo $C(y)=0$, si ha

$U_1(x,y) = \frac{1}{2} \log((x-1)^2 + y^2)$, potenziale per w_1 su $\text{dom}(w_1)$ [e quindi su $\text{dom}(w)$].

$\Rightarrow w_1$ esatta in $\text{dom}(w)$.

•) Studiamo w_2 .

$$\frac{\partial w_{2,x}}{\partial y} = \frac{\partial w_{2,y}}{\partial x} \longrightarrow w_2 \text{ chiusa in } \text{dom}(w_2) \supset \text{dom}(w)$$

$\text{dom } w_2$ non è simplic. connesso, quindi non so nulla sull'esattezza: in effetti non è esatta globalmente sul suo dominio. Vediamolo.

$$u_2(x,y) = \int w_{2,x}(x,y) dx = \arctan\left(\frac{x+1}{y}\right) + D(y)$$

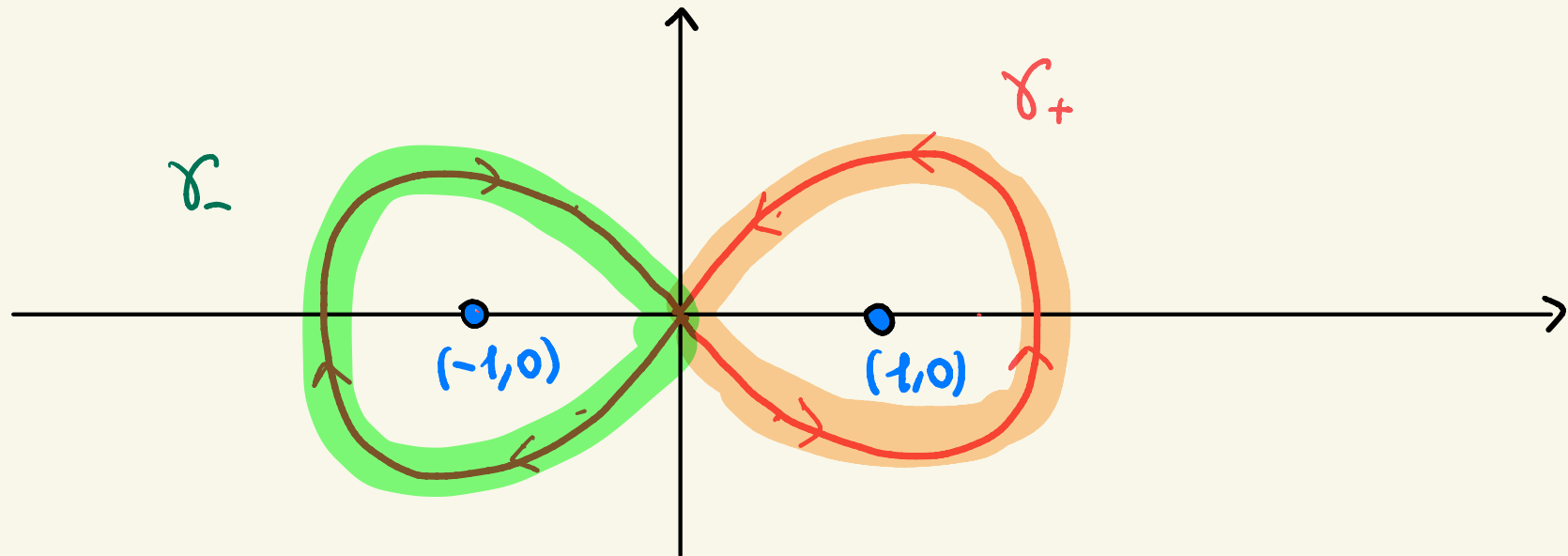
Imponiamo $\frac{\partial u_2}{\partial y} = w_{2,y}$, troviamo $D'(y) = 0$.

Scego $D(y) = -\pi/2$, così $u_2(x,y) = \arctan\left(\frac{x+1}{y}\right) - \pi/2$, potenziale per w_2 su $\{y > 0\}$ e $\{y < 0\}$.

Ma, $\forall t, \pi/2 - \arctan(1/t) = \arctan t$, dunque:

$\tilde{u}_2(x,y) = -\arctan\left(\frac{y}{x+1}\right)$ potenziale per w_2 su $\{x > -1\}$ (o separatamente su $\{x < -1\}$).

$\Rightarrow w_2$ è esatta in $\{x > -1\}$ (o in $\{x < -1\}$) ma non in dom w



$$\int_{\gamma} \omega = \underbrace{\int_{\gamma} \omega_1}_{=0} + \int_{\gamma} \omega_2 + \int_{\gamma} \hat{\omega}_{+1} = \int_{\gamma} \omega_2 + \int_{\gamma} \hat{\omega}_{+1}.$$

perché γ chiusa
e ω_1 esatta

Scriviamo γ come unione di due curve

$$\gamma_+(\theta) = (2\cos\theta, \sin(2\theta)) \quad \theta \in [-\pi/2, \pi/2]$$

$$\gamma_-(\phi) = (2\cos\phi, \sin(2\phi)) \quad \phi \in [\pi/2, 3/2\pi]$$

$$\Rightarrow \int_{\gamma} \omega = \int_{\gamma_+} \omega_2 + \int_{\gamma_-} \omega_2 + \int_{\gamma_+} \hat{\omega}_{+1} + \int_{\gamma_-} \hat{\omega}_{+1}$$

ω_2 e $\hat{\omega}_{+1}$ chiuse $\Rightarrow$ USO INVARIANZA OMOTOPICA

$\gamma_+ \sim \ell_+(t) = (1 + \cos t, \sin t)$, $t \in [0, 2\pi]$ (circonf. raggio 1, centro (1,0), senso orario)

$\gamma_- \sim \ell_-(s) = (-1 + \cos s, -\sin s)$, $s \in [0, 2\pi]$ (circonf. raggio 1, centro (-1,0), senso antiorario)

$$\ell'_+(t) = (-\sin t, \cos t)$$

$$\ell'_-(s) = (-\sin s, -\cos s)$$

- ω_2 esatta su $\{x > -1\}$ (perché $\{x > -1\} \cap \text{dom}(\omega_2)$ è sempl. conn.) $\Rightarrow \int_{\gamma_+} \omega_2 = 0$
- $\hat{\omega}_{+1}$ esatta su $\{x < 1\}$ (perché $\{x < 1\} \cap \text{dom}(\hat{\omega}_{+1})$ è sempl. conn.) $\Rightarrow \int_{\gamma_-} \hat{\omega}_{+1} = 0$

$$\Rightarrow \int_{\gamma} \omega = \int_{\gamma^-} \omega_2 + \int_{\gamma^+} \hat{\omega}_{+1} = 2\pi + 2\pi = 4.$$

ESERCIZIO 3

i) $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2 - \frac{4}{3}|y - \frac{5}{2}|\}$

$$w(x, y) = (y^2 e^x - 7y) dx + (2ye^x - 7) dy.$$

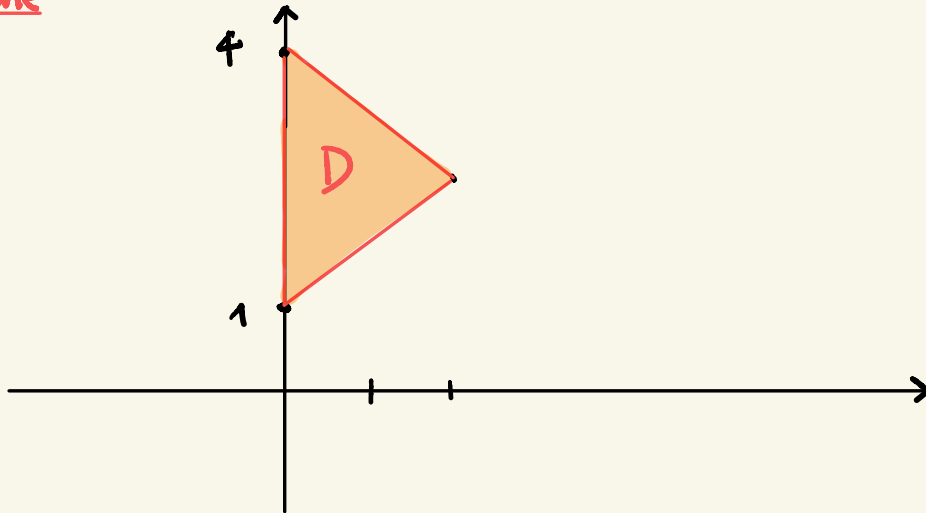
Calcolare $\int_{\partial D} w$.

ii) Sia T il toro del testo (vedi foglio), ed $F = (x^2 y, -xy^2, z)$.

Calcolare il flusso di F uscente da T .

Soluzione

i)



$$w = w_1 + w_2$$

con $w_1 = y^2 e^x dx + (2ye^x - 7) dy$

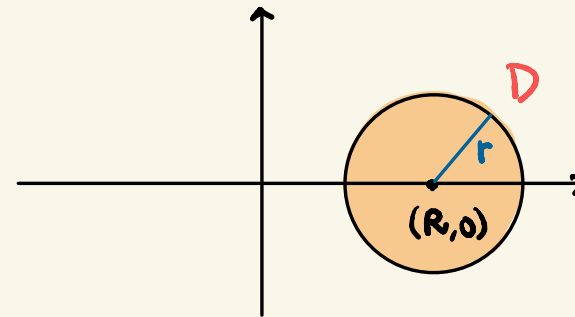
$$w_2 = -7y dx$$

• w_1 chiusa su $\mathbb{R}^2 \Rightarrow$ esatta $\Rightarrow \int_{+\partial D} w_1 = 0$.

• $\int_{+\partial D} w_2 = -7 \int_{+\partial D} y dx \stackrel{\text{Gours-Green}}{=} 7 \iint_D dx dy = 7 \text{Area}(D) = 7 \frac{3 \cdot 2}{2} = 21$.

ii) $\int_T F \cdot \vec{n}_{\text{ext}} d\sigma \stackrel{\text{Th. Divergenza}}{=} \iiint_T \text{div} F dx dy dz = \text{Vol}(\text{Toro Pieno})$.

$\text{Vol}(T) = 2\pi R \text{Area}(D) = 2\pi R \pi r^2 = 2\pi^2 r^2 R$
Pappo-Guldino



ESERCIZIO 4

$$E(x, y, z) = A(x, y, z) \cdot (x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z})$$

$$\text{con } A(x, y, z) = \frac{q}{(4\pi\epsilon_0)} r^{-3/2} (1 + r^{1/2}/\lambda) e^{-r^{1/2}/\lambda}$$

$$r = r(x, y, z) = x^2 + y^2 + z^2$$

$$\lambda > 0$$

Si ponga $q/(4\pi\epsilon_0) = 1$ così da avere $q/\epsilon_0 = 4\pi$.

Sia $S_R = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq R^2\}$, $R > 0$.

Mostrare che

$$\iint_{\partial S_R} \langle E, \nu_{\text{ext}} \rangle d\sigma + \frac{1}{\lambda^2} \iiint_{S_R} V(x, y, z) dx dy dz = 4\pi = \frac{q}{\epsilon_0}$$

con V tale che $E = -\nabla V$.

Soluzione

$\text{rot}(E) = 0$ (calcolarlo!)

$\text{dom}(E) = \mathbb{R}^3 \setminus \{(0, 0, 0)\} \implies \text{simpl. connesso} \implies E$ conservativo.

Calcoliamo il potenziale V .

$$-V = \int E^x(x, y, z) dx = \int A \cdot x dx$$

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2 + z^2} \\ y, z &\text{ const.} \\ \rho^2 &= x^2 + y^2 + z^2 \\ \rho d\rho &= x dx \end{aligned}$$

parti sul
secondo $\int$: $f(\rho) = \frac{1}{\rho^{-\rho/\lambda}} \rightarrow f'(\rho) = -\frac{1}{\rho^2}$
 $g(\rho) = e^{-\rho/\lambda} \rightarrow g'(\rho) = -\lambda e^{-\rho/\lambda}$

$$\begin{aligned} &= \int A(\rho) \rho d\rho \\ &= \int \frac{1}{\rho^3} (1 + \rho/\lambda) e^{-\rho/\lambda} \rho d\rho \\ &= \int \frac{e^{-\rho/\lambda}}{\rho^2} d\rho + \frac{1}{\lambda} \int \frac{e^{-\rho/\lambda}}{\rho} d\rho \\ &= \int \frac{e^{-\rho/\lambda}}{\rho^2} d\rho + \frac{1}{\lambda} \left(-\frac{1}{\rho} e^{-\rho/\lambda} \right) - \frac{1}{\lambda} \left(\int -\frac{1}{\rho^2} (-\lambda e^{-\rho/\lambda}) d\rho \right) \\ &= \int \frac{e^{-\rho/\lambda}}{\rho^2} d\rho - \frac{e^{-\rho/\lambda}}{\rho} - \int \frac{e^{-\rho/\lambda}}{\rho} d\rho + C(y, z) \\ &= \frac{e^{-\sqrt{x^2 + y^2 + z^2}/\lambda}}{\sqrt{x^2 + y^2 + z^2}} + C(y, z) \end{aligned}$$

Ripetiamo il ragionamento per F^y e F^z . Otteniamo:

$$V(x, y, z) = \frac{e^{-\sqrt{x^2 + y^2 + z^2}/\lambda}}{\sqrt{x^2 + y^2 + z^2}} + C, \quad C \in \mathbb{R}.$$

Poniamo $C=0$.

$$V(x, y, z) = \frac{e^{-\sqrt{x^2 + y^2 + z^2}/\lambda}}{\sqrt{x^2 + y^2 + z^2}}$$

Ora:

$$\partial S_R = \Psi(D) \quad \text{con} \quad D = ([0, \pi] \times [0, 2\pi])$$

$$\Psi(\theta, \varphi) = (R \sin \theta \cos \varphi, R \sin \theta \sin \varphi, R \cos \theta)$$

$$\Psi_\theta \wedge \Psi_\varphi = R^2 \sin^2 \theta \cos \varphi \, \underline{i} + R^2 \sin^2 \theta \sin \varphi \, \underline{j} + R^2 \cos \theta \sin \theta \, \underline{k}$$

$$\begin{aligned} \text{Allora: } \iint_{\partial S_R} \langle \vec{E}, \vec{\nu}_{\text{ext}} \rangle d\sigma &= \int_0^\pi \int_0^{2\pi} A(R) R^2 \left\langle \begin{pmatrix} R \sin \theta \cos \varphi \\ R \sin \theta \sin \varphi \\ R \cos \theta \end{pmatrix}, \begin{pmatrix} \sin^2 \theta \cos \varphi \\ \sin^2 \theta \sin \varphi \\ \cos \theta \sin \theta \end{pmatrix} \right\rangle d\varphi d\theta \\ &= R^3 A(R) \cdot 2\pi \int_0^\pi (\sin^3 \theta + \cos^2 \theta \sin \theta) d\theta \\ &= 4\pi R^3 A(R) \end{aligned}$$

$$\begin{aligned} \text{Inoltre: } \iiint_{S_R} V(x, y, z) dx dy dz &= \int_0^{2\pi} d\varphi \cdot \int_0^\pi \sin \theta d\theta \int_0^R \rho e^{-\rho/\lambda} d\rho \\ &= 4\pi \int_0^R \rho e^{-\rho/\lambda} d\rho \\ \text{Part: } \begin{matrix} f(\rho) = \rho \rightarrow f'(\rho) = 1 \\ g'(\rho) = e^{-\rho/\lambda} \rightarrow g(\rho) = -\lambda e^{-\rho/\lambda} \end{matrix} &= 4\pi \left(-\lambda [\rho e^{-\rho/\lambda}]_0^R + \lambda \int_0^R e^{-\rho/\lambda} d\rho \right) \\ &= 4\pi \left(-\lambda R e^{-R/\lambda} - \lambda^2 e^{-R/\lambda} + \lambda^2 \right) \end{aligned}$$

Concludiamo

$$\begin{aligned} \iint_{\partial S_R} \langle \vec{E}, \vec{\nu}_{\text{ext}} \rangle d\sigma + \frac{1}{\lambda^2} \iiint_{S_R} V(x, y, z) dx dy dz &= 4\pi \left(R^3 A(R) - \frac{R}{\lambda} e^{-R/\lambda} - e^{-R/\lambda} + 1 \right) \\ A(R) = \frac{1}{R^3} \left(1 + \frac{R}{\lambda} \right) e^{-R/\lambda} &= 4\pi \left[\left(1 + \frac{R}{\lambda} \right) e^{-R/\lambda} - \left(1 + \frac{R}{\lambda} \right) e^{-R/\lambda} + 1 \right] \\ &= 4\pi \\ 1 = q/4\pi \epsilon_0 &= q/\epsilon_0 \end{aligned}$$